Jurong Pioneer Junior College

2026 H2 Computing (Syllabus 9569)

Test 2 Revision

1 JP canteen operates various food stalls in school and wishes to implement a web-based food ordering system to better understand students' food taste and facilitate food ordering. Students can use the food ordering system to browse the dishes, place their orders on dishes and collect when their food is ready.

The web-based food ordering system must be able to:

- Store student details.
- Store information related to the stalls and the dishes sold by them.
- Keep track of the dishes ordered by students.

There are a total of **four** tables, and the table descriptions are as follows:

- `student(studentID, name, phoneNo)`
- `stall(stallID, stallName)`
- `dish(dishID, stallID*, dishName, price, availability)`
- `orderDish(studentID*, dishID*, orderTime, orderDate, quantity)`

The primary key is indicated by underlining one or more attributes.
The attribute indicated with * denoted a foreign key.

It is given that:

- students are allowed to order different dishes in a single transaction. For example, when a student orders **three** different dishes of **one** item in a single transaction, **three** separate records with the same value for `studentID`, `orderTime`, `orderDate` and `quantity` will be inserted into the table `orderDish`
- the primary keys `stallID` and `dishID` of the tables `stall` and `dish` are auto-incremented integers.
- `orderDate` and `orderTime` in table `orderDish` are represented in the format of `YYYY-MM-DD` and `HH:MM` respectively.

Task 1.1

Create a SQL file called

`TASK1_1_<your_name>_<centre_number>_<index_number>.sql` to show the SQL code to create database `canteen.db` with the **four** tables given. Define the primary and foreign keys for each table.

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Task 1.2

The files `STUDENT.txt`, `STALL.txt`, `DISH.txt` and `ORDERDISH.txt` contain information about the students, stalls, dishes and order history respectively for insertion into the canteen database. Each row in the four files is a comma-separated list of information.

FOR `STUDENT.txt`, information about each student is given in the following order:

`NRIC`, `Name`, `PhoneNo`

Due to the sensitive nature of the National Registration Identity Card (NRIC), and how it can be subjected used to access a wide range of other personal data. it is decided that a unique identifier for each student is to be obtained by combining the last four characters of NRIC and the first four characters of a student's name with white spaces removed. All letters in `studentID` will be converted to uppercase.

For example, a student with `NRIC = "S2345678B"` and `name = "Low Jia Ying"` will be given a `studentID` of "`LOWJ678B` ".

For `STALL.txt`, information about each stall is given in the following order:

`stallID`, `stallName`.

For `DISH.txt,` information about each dish is given in the following order:

`dishID`, `stallID`, `dishName`, `price` and `availability`.

For `ORDERDISH.txt`, information about the dishes ordered by individual students over time

Write a Python program to insert all information from the **four** files into the **four** separate tables of the canteen database, `canteen.db` based on the table descriptions given above. Run the program.

Save your program code as

`Task1_2_<your name>_<centre number>_<index number>.py`

Task 1.3

Write SQL code to show the dish names and prices of all the **available** dishes with a specified stall name. Run this query with the store name "`Spicy Delight`".

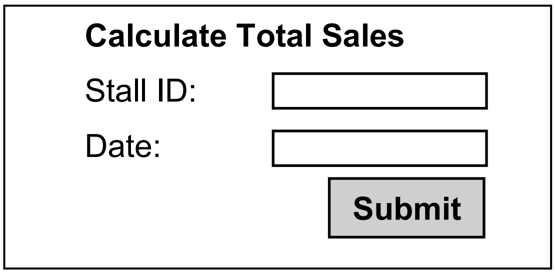
Save this code as.

`Task1_3_<your name>_<centre number>_<index number>.sql` [10]

Task 1.4

The stall owners of JP Canteen requested that they would like to know their total sales amount collected on a particular date.

Write a Python program and the necessary files to create a web application. The web application accepts **two** input Stall ID and Date from the web form in separate textboxes. The form design of the web application is shown below:



Save your program code as

`Task1_4_<your name>_<centre number>_<index number>.py`

With any additional files/subfolders as needed in a folder named

`Task1_4_<your name>_<centre number>_<index number>`

Run the web application and save the output of the program as [4]

`Task1_4_<your name>_<centre number>_<index number>.html`

Task 1.5

Write an SQL query that sums up the total sales amount collected by a stall specified by `stallID` on a given `date`.

The results of the query should be shown on a webpage in a table that displays the following data:

- Stall ID
- Stall Name
- Date
- Total Sales Collected

The program should return a HTML document that enables the web browser to display a table with required data. The resulting web page is accessible by clicking on the "Submit" button on the web form in Task 1.4.

The program should return an HTML document that enables the web browser to display a table with the required data.

Save your program code as

`Task1_5_<your name>_<centre number>_<index number>.py`

With any additional files/subfolders as needed in a folder named

`Task1_5_<your name>_<centre number>_<index number>`

Run the web application and save the output of the program as [4]

`Task1_5_<your name>_<centre number>_<index number>.html`

Task 1.6

Run your web application by entering the following input into the web form through its "Submit" button.

- `Stall ID` = "`Nasi Padang`"
- `Date` = "`2025-07-04`"

Run the web application and save the output of the program as [3]

`Task1_6_<your name>_<centre number>_<index number>.html`